Morphisms of Directed Containers as Cofunctors

Extracted from a machine-checked development

Abstract

A directed container presents a small category whose objects are shapes and whose morphisms are positions. Morphisms of directed containers—shape maps forward, position maps *backward*, compatible with root, sub-shape, and shift—are not functors of the presented categories: the variance on hom-sets is opposite. They are precisely *cofunctors* (retrofunctors). We show that the two structures carry identical data, so there is a bijection on hom-sets between directed-container morphisms and cofunctors; that cofunctors form a strict category; and that the correspondence preserves identities and composition, yielding an isomorphism of categories **DCont** $\cong$ **Cof**. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

A directed container equips a container (S, P) with a root $\mathsf{o}(s)$, a sub-shape $s{\downarrow}p$, and a shift $p \oplus q$, subject to laws D1–D5. Reading these operations categorically, shapes are objects and positions out of a shape are morphisms; the root is the identity and the shift is composition. This is the object-level dictionary of Ahman–Uustalu: directed containers present small categories.

At the morphism level a tension appears. A container morphism has a *covariant* shape map and a *contravariant* position map. A functor between the presented categories would act covariantly on hom-sets. The two variances are opposite, so a directed-container morphism cannot be a functor. With compatibility laws for root, sub-shape, and shift, it is exactly a **cofunctor**: an object map together with a backward lift of morphisms.

What this note claims. In ordinary mathematical language: the data of a directed-container morphism and the data of a cofunctor between the presented categories are the same data, under the dictionary that renames root/sub/shift to identity/codomain/ composition. Consequently there is a hom-set bijection; cofunctors compose strictly; and the bijection is an isomorphism of categories $\mathbf{DCont} \cong \mathbf{Cof}$.

2 Directed containers and the presented category

We work in a single universe **Type**. A **directed container** D is a container $(\mathrm{Shape}(D), \mathrm{Pos}(D))$ with root, sub-shape, and shift operations satisfying the five directed laws D1–D5 (as in the companion note on directed containers and the induced comonad). We write S and P for shapes and positions when convenient.

Through the categorical dictionary we introduce the following vocabulary.

Definition 2.1 (Dictionary)**.** Let D be a directed container.

1. The **identity morphism** at a shape s is the root: $\mathrm{id}_s := \mathsf{o}(s) \in P(s)$.
2. The **codomain** of a position $p \in P(s)$ is the sub-shape: $\mathrm{cod}(s, p) := s \downarrow p \in S$.
3. The **composition** of $p \in P(s)$ with $q \in P(s \downarrow p)$ is the shift: $q \circ p := p \oplus q \in P(s)$ (diagrammatic order: travel along p, then along q).

Remark 2.2. In this note we treat the presented category through this dictionary without re-deriving the category laws; those laws are D2/D3 (units) and D5 (associativity), transported along D1 and D4, and are established in the companion development on directed containers as categories.

3 Morphisms of directed containers

Definition 3.1 (Directed-container morphism)**.** Let C and D be directed containers. A **morphism of directed containers** $\varphi : C \Rightarrow D$ consists of

1. a map on shapes $f : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$, and
2. for each $s \in \mathrm{Shape}(C)$, a function $f^\sharp_s : \mathrm{Pos}(D)(f(s)) \to \mathrm{Pos}(C)(s)$ (contravariant on positions),

subject to the three compatibility laws M0–M2 below.

Definition 3.2 (Laws M0–M2)**.** Let $\varphi = (f, f^\sharp) : C \Rightarrow D$ be as above.

M0 (codomain). For all $s \in \mathrm{Shape}(C)$ and $h \in \mathrm{Pos}(D)\big(f(s)\big)$,

$$f\big(s \downarrow f^\sharp_s(h)\big) \;=\; f(s) \downarrow h.$$

M1 (unit). For all $s \in \mathrm{Shape}(C)$,

$$f^\sharp_s\big(\mathsf{o}(f(s))\big) \;=\; \mathsf{o}(s).$$

M2 (composition). For all $s \in \mathrm{Shape}(C)$, $h \in \mathrm{Pos}(D)\big(f(s)\big)$, and $k \in \mathrm{Pos}(D)\big(f(s) \downarrow h\big)$,

$$f^\sharp_s(h \oplus k) \;=\; f^\sharp_s(h) \;\oplus\; f^\sharp_{s \downarrow f^\sharp_s(h)}\big((\mathrm{M0}_{s,h})^{-1} \cdot k\big),$$

where the inner position k is transported along M0 so that the shift in C is well-typed.

Remark 3.3. Law M2 transports along M0 exactly as the directed law D5 transports along D4: the right factor of the shift lives in a fibre identified only up to a shape equality.

4 Cofunctors

Definition 4.1 (Cofunctor)**.** Let C and D be directed containers, and write $\mathcal{C}(C)$ and $\mathcal{C}(D)$ for the categories they present. A **cofunctor** $\phi : \mathcal{C}(C) \to \mathcal{C}(D)$ consists of

1. an object map $\mathrm{obj} : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$, and
2. for each $s \in \mathrm{Shape}(C)$, a **lift** $\mathrm{lift}_s : \mathrm{Pos}(D)(\mathrm{obj}(s)) \to \mathrm{Pos}(C)(s)$ sending a morphism out of $\mathrm{obj}(s)$ to a morphism out of s,

subject to the three cofunctor laws C0–C2 below.

Definition 4.2 (Laws C0–C2)**.** Let $\phi = (\mathrm{obj}, \mathrm{lift})$ be as above.

C0 (codomain). For all s and $h \in \mathrm{Pos}(D)(\mathrm{obj}(s))$,

$$\mathrm{obj}\big(\mathrm{cod}(s,\, \mathrm{lift}_s(h))\big) \;=\; \mathrm{cod}\big(\mathrm{obj}(s),\, h\big).$$

C1 (unit). For all s,

$$\mathrm{lift}_s\big(\mathrm{id}_{\mathrm{obj}(s)}\big) \;=\; \mathrm{id}_s.$$

C2 (composition). For all s, h, and k of the appropriate types,

$$\mathrm{lift}_s(k \circ h) \;=\; \mathrm{lift}_s(h) \;\circ\; \mathrm{lift}_{\mathrm{cod}(s,\mathrm{lift}_s(h))}\big((\mathrm{C0}_{s,h})^{-1} \cdot k\big),$$

with transport of k along C0 as in M2.

Remark 4.3 (Variance)**.** A functor $\mathcal{C}(C) \to \mathcal{C}(D)$ would supply, for each s, a map $\mathrm{Pos}(C)(s) \to \mathrm{Pos}(D)(F(s))$—covariant on hom-sets. A cofunctor supplies the opposite map. The object map still runs $C \to D$; only the action on morphisms is reversed. This is the variance forced by container morphisms.

5 The hom-set bijection

Under the dictionary of Definition 2.1, the data and laws of a directed-container morphism are literally those of a cofunctor: f is obj, $f^\sharp$ is lift, and M0/M1/M2 are C0/C1/C2.

Definition 5.1 (From morphisms to cofunctors)**.** Every directed-container morphism $\varphi = (f, f^\sharp) : C \Rightarrow D$ determines a cofunctor $\varphi^\wedge : \mathcal{C}(C) \to \mathcal{C}(D)$ with $\mathrm{obj} = f$ and $\mathrm{lift} = f^\sharp$.

Definition 5.2 (From cofunctors to morphisms)**.** Every cofunctor $\phi = (\mathrm{obj}, \mathrm{lift})$ determines a directed-container morphism $\phi^\vee : C \Rightarrow D$ with shape map obj and position map lift.

Theorem 5.3 (Round-trip, morphism side)**.** *For every directed-container morphism φ,*

$$\big(\varphi^\wedge\big)^\vee \;=\; \varphi. \tag{1}$$

Proof. By lean. □

Theorem 5.4 (Round-trip, cofunctor side)**.** *For every cofunctor ϕ,*

$$\big(\phi^\vee\big)^\wedge \;=\; \phi. \tag{2}$$

Proof. By lean. □

Corollary 5.5 (Hom-set bijection)**.** *For any directed containers C and D, there is a bijection*

$$\{\ \textit{directed-container morphisms } C \Rightarrow D\ \} \ \cong\ \{\ \textit{cofunctors } \mathcal{C}(C) \to \mathcal{C}(D)\ \}, \tag{3}$$

implemented by $(\,\cdot\,)^{\wedge}$ and $(\,\cdot\,)^{\vee}$.

Proof. By lean. (The two constructions are mutually inverse by Theorems 5.3 and 5.4.) □

Remark 5.6. Both round-trips are definitional: the two structures carry identical fields, and the laws are the same equalities under renaming. No computation is required beyond unfolding definitions.

6 The category of cofunctors

Definition 6.1 (Identity cofunctor)**.** The **identity cofunctor** on $\mathcal{C}(C)$ has object map the identity on shapes and lift, at each shape, the identity on positions. The laws C0–C2 hold strictly (transports along reflexivity vanish).

Definition 6.2 (Composition of cofunctors)**.** Let $\phi : \mathcal{C}(C) \to \mathcal{C}(D)$ and $\psi : \mathcal{C}(D) \to \mathcal{C}(E)$ be cofunctors. Their **composite** $\psi \circ \phi : \mathcal{C}(C) \to \mathcal{C}(E)$ has

1. object map $\mathrm{obj}_{\psi} \circ \mathrm{obj}_{\phi}$, and
2. lift at s given by
$$m \ \mapsto\ \mathrm{lift}_{\phi}(s,\, \mathrm{lift}_{\psi}(\mathrm{obj}_{\phi}(s),\, m))$$
(lifts compose backwards).

The composite codomain law chains the two factors' C0 laws; the composite unit law chains the two C1 laws; the composite composition law uses the factors' C2 laws together with a transport identity recorded next.

Lemma 6.3 (Composite transport)**.** *Let $\psi : \mathcal{C}(D) \to \mathcal{C}(E)$ be a cofunctor, let $x, y \in \mathrm{Shape}(D)$ with $e : y = x$, and let $z \in \mathrm{Shape}(E)$ with $e_2 : \mathrm{obj}_{\psi}(x) = z$. For $k \in \mathrm{Pos}(E)(z)$,*

$$e^{-1} \cdot \big(\mathrm{lift}_{\psi}(x,\, e_2^{-1} \cdot k)\big) \ =\ \mathrm{lift}_{\psi}\big(y,\, ((\mathrm{obj}_{\psi} \cdot e) \cdot e_2)^{-1} \cdot k\big). \tag{4}$$

Proof. By lean. □

Remark 6.4. This is the one genuinely dependent step in the composition of cofunctors: the inner position, transported across both factors' C0 equalities in one hop, equals the two-hop transport. After substituting the shape equalities the identity is reflexivity.

Proposition 6.5 (Left unit for cofunctors)**.** *For every cofunctor $\phi : \mathcal{C}(C) \to \mathcal{C}(D)$,*

$$\phi \ \circ\ \mathrm{id}_{\mathcal{C}(C)} \ =\ \phi. \tag{5}$$

Proof. By lean. □

Proposition 6.6 (Right unit for cofunctors)**.** *For every cofunctor $\phi : \mathcal{C}(C) \to \mathcal{C}(D)$,*

$$\mathrm{id}_{\mathcal{C}(D)} \ \circ\ \phi \ =\ \phi. \tag{6}$$

Proof. By lean. □

Proposition 6.7 (Associativity for cofunctors)**.** *Composition of cofunctors is associative:*

$$\chi \circ (\psi \circ \phi) \;=\; (\chi \circ \psi) \circ \phi. \tag{7}$$

Proof. By lean. □

Theorem 6.8 (The category **Cof**)**.** *Directed containers and cofunctors form a strict category* **Cof***: identities and composition as above, with unit and associativity laws holding on the nose.*

Proof. By lean. (Propositions 6.5, 6.6, and 6.7.) □

7 Isomorphism of categories

Transporting the identity cofunctor and cofunctor composition across the hom-set bijection yields identity and composition for directed-container morphisms.

Definition 7.1 (Identity and composition in **DCont**)**.** The **identity** directed-container morphism on C is $(\mathrm{id}_{\mathcal{C}(C)})^{\vee}$. The **composite** of morphisms $\varphi : C \Rightarrow D$ and $\psi : D \Rightarrow E$ is $\left(\psi^{\wedge} \circ \varphi^{\wedge}\right)^{\vee}$.

Proposition 7.2 (Preservation of identities)**.** *The assignment $\Phi(\varphi) = \varphi^{\wedge}$ preserves identities:*

$$(\mathrm{id}_C)^{\wedge} \;=\; \mathrm{id}_{\mathcal{C}(C)}. \tag{8}$$

Proof. By lean. □

Proposition 7.3 (Preservation of composition)**.** *The assignment $\Phi(\varphi) = \varphi^{\wedge}$ preserves composition:*

$$(\psi \circ \varphi)^{\wedge} \;=\; \psi^{\wedge} \circ \varphi^{\wedge}. \tag{9}$$

Proof. By lean. □

Theorem 7.4 (Isomorphism of categories)**.** *The assignment Φ is an isomorphism of categories*

$$\mathbf{DCont} \;\cong\; \mathbf{Cof}. \tag{10}$$

It is the identity on objects, bijective on every hom-set (Corollary 5.5), and functorial (Propositions 7.2 and 7.3).

Proof. By lean. □

8 Discussion

What has been established. Directed-container morphisms and cofunctors of the presented categories are the same mathematical objects under a change of vocabulary. The category they form is strict, and the identification is an isomorphism of categories. In particular, there are no exotic directed-container morphisms beyond cofunctors, and no exotic cofunctors beyond directed-container morphisms.

Why not functors? The position component of a container morphism is contravariant by the semantics of extensions: positions of the target are pulled back to positions of the source so that labellings can be reindexed. Functors act covariantly on hom-sets. The mismatch is definitional, not accidental; cofunctors are the correct dual notion.

Scope. This note takes the object-level dictionary (directed containers as categories) as given and focuses on the morphism-level identification. The full round-trip at the object level—recovering a directed container from its presented category up to isomorphism of positions—belongs to the companion note on directed containers as categories.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content follows the directed-container theory of Ahman, Chapman, and Uustalu, and the cofunctor perspective associated with the Shapiro–Spivak polynomial dictionary. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

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